Festus Basimtaal Ayembilla

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Education

Chungbuk National University South Korea.
MSc Architectural Engineering (CGPA 3.78/4.0) September 2024 - August 2026
Relevant Coursework: Advanced Architectural IT Programming, Architectural IT Programming Fundamentals.

University of Energy and Natural Resources (UENR) Ghana.
BSc Civil Engineering (CGPA 3.14/4.0) September 2018 – October 2022
Relevant Coursework: Systems Engineering, Structural Engineering, Computer Programming for Civil Engineers

Research Interests

- Scan-to-BIM, Construction Robotics, Multi-Robot Coordination for Automated Construction, Digital Twins, 3D Point Cloud Processing, Computer Vision for Construction.

Publications and Manuscripts

- **AYEMBILLA, F. B.,** RAJENDRAN, M., & Min-Koo, K. I. M. (2025). Improving Generalization of Deep Learning Models in MEP Point Cloud Segmentation Through Synthetic Data Augmentation. In International conference on construction engineering and project management (pp. 306-313). Korea Institute of Construction Engineering and Management. https://doi.org/10.6106/ICCEPM.2025.0306.
- Dao H., **Ayembilla, F.B**., Sim, S., Won, J., Kang, M., Yoon, S., Kim, H., Kim M. (2025) Completing occluded indoor point clouds for enhanced semantic segmentation in Scan-to-BIM workflows. FITAT 2025 Ho Chi Minh, Vietnam. **(Accepted)**
- **AYEMBILLA, F. B.,** KIM, T., KIM, M. K., Seo, J., & KIM, J. I. An Automated Pipeline for Generating a Multi-Layered Spatial Map for Indoor Robot Navigation from BIM Data. representations, 16, 17. https://doi.org/10.22260/ISARC2026/0030.
- Mathanraj Rajendran, **Ayembilla Festus Basimtaal**, Sung-Han Sim, Min-Koo Kim, and Chang-Yong Yi, Automated Shape-Aware Dimensional Classification and Semantic-Instance Segmentation of Steel Structures from Occluded Point Cloud Data, Submitted to Automation in Construction, 2026. https://doi.org/10.1016/j.autcon.2026.107241
- Rajendran, M., **Ayembilla, F.B.**, Sim, S. H., & Kim, M. K. Geometric-Based Dimensional Classification of Steel Structures with Semantic and Instance Segmentation from 3D Point Cloud Data. In International Conference on Computing in Civil Engineering (i3CE 2026, South Korea). **(Accepted)**.
- **Ayembilla F. B.**, Kim, M.K. (2026). BIM-PT: A Hybrid Pipeline with Cross-Area Structural Augmentation for Indoor Point Cloud Semantic Segmentation. CIB W78 2026 International Conference on IT in Construction. **(Accepted).**
- **AYEMBILLA, F. B**., & Kim, M. K. (2026). Improving Cross-Facility Generalization of MEP Semantic Segmentation Using Independent-Source Synthetic Point Clouds. Advanced Engineering Infomatics. **(Under review)**

Research Experience

Graduate Research Assistant
Smart Constructions and Systems Laboratory, Chungbuk National University September 2024 - Present
Supervisor: Professor Minkoo Kim

- Investigating advanced engineering technologies for improving Scan-to-BIM and facility management in the construction industry.
- Designing a Blender-based TLS simulation framework to generate occlusion-aware MEP point clouds with

configurable beam divergence, incident angle, and realistic noise, exporting XYZ, normals, and semantic labels.

- Collecting 3D data from Terrestrial and Mobile Laser Scanners (TLS and MLS) to support experiments in structural inspection and indoor scene reconstruction. The collected data are preprocessed and organized into standardized datasets for segmentation and dimensional analysis tasks.
- Investigating the effects of synthetic data generation and geometry completion, using Generative Adversarial Networks (GANs) and diffusion models, on the segmentation accuracy of deep learning architectures such as ResPointNet++.
- Developing a multi-layer spatial map schema and unified absolute coordinate system that enables heterogeneous industrial robots to share consistent localization and mapping data for autonomous indoor navigation and Scan-to-BIM integration.

Graduate Research Assistant

Department of Civil and Environmental Engineering, UENR October 2022 – December 2023

- Conducted experimental testing to determine the tensile strength, failure strain, and modulus of elasticity for GFRP bars with diameters of 10 mm, 12 mm, and 16 mm under varying loading conditions. Compared results against conventional steel bars to assess structural viability and mechanical performance.
- Assisted students in the design and execution of laboratory and field experiments, ensuring accurate data collection through standardized procedures and quality control protocols.
- Analyzed experimental data using SPSS, Excel, MATLAB, and ANSYS to extract mechanical and structural insights relevant to composite material performance.

Projects

Robot-Adaptive Building (Core Technologies for Design/Construction/Operations Management).

Supervisor: Professor Minkoo Kim July 2025 – Present

Scope: Developing frameworks that enable multiple robots to operate within a shared, BIM-anchored spatial environment for construction and facility management.

Key Contributions:

- Investigating automated scan-to-BIM approaches for reconstructing point cloud data to 3D BIM models.
- Designed a Universal Coordinate System (UCS) for linking BIM, real-world, and robot coordinate frames, defining unit standards, axis conventions, and anchor placement rules.
- Created a multi-layer spatial map model integrating geometry, topology, and semantic information with version control and update mechanisms for real-time robot use.
- Built an IFC-to-robot conversion pipeline that generates 3D and 2D maps (GLB, JSON, PGM/YAML) compatible with ROS2 navigation and visualization tools.

50-unit apartment complex for lecturers of the University of Energy and Natural Resources.

Supervisor: Dr Nana Ackerson January 2022 - April 2022

- Utilized Autodesk Revit, AutoCAD Architecture, and Lumion 12.0 to design and prepare architectural and structural drawings for an estate comprising 50 residential two-bedroom apartments, one clinic, one supermarket, two basketball and volleyball courts, and a football park. Project Link

Teaching Experience

Department of Civil Engineering, Bolgatanga Technical University Bolgatanga, Ghana

BTEC 418: Advanced Design (***Enrolment - 15***) Spring 2024

- Graded laboratory reports, homework assignments, quizzes, and midterm exams, providing constructive feedback to support student learning.

- Guided students in the application of structural design principles using AutoCAD and Revit.
- Assisted in preparing instructional materials and demonstrating design workflows.

BTEC 314: Structural Analysis II (***Enrolment - 16***) Spring 2024

- Conducted tutorials on statically indeterminate structures and influence line analysis.
- Evaluated student problem sets and clarified complex analysis procedures during lab sessions.

BTC 323: Structural Design I (***Enrolment - 16***) Summer 2024

- Conducted tutorials in reinforced concrete and steel design principles.
- Provided academic support and assisted with grading assignments.

BTC 224 Structural Analysis I (***Enrolment - 10***) Summer 2024

- Prepared supplementary learning materials and conducted tutorials in methods of joint, section, and deflection analysis.

Department of Civil Engineering, UENR Sunyani, Ghana

CIVL 104 Introduction to Civil Engineering Drawing Spring 2023

- As the primary instructor, I prepared and organized course materials, including slides, handouts, and lab manuals, and managed content on the university's learning management system.
- Taught lectures, lab sessions, tutorials, and reviewed assignments.

Work Experience

GEOMYK Construction and Engineering LTD Sunyani – Ghana

Part-Time Engineer January 2023 – September 2024

- Prepared architectural and structural documents for construction
- Performed on-site supervision of construction works
- Prepared project schedules and construction project reports

Ghana Highway Authority Sunyani – Ghana

Intern October 2021 – December 2021

- Participated in laboratory experiments on soil samples, coarse aggregates, and concrete cubes for site projects.
- Visited construction sites to interact with site engineers and understand the workflow on site.

Architectural Engineering Service Limited Bolgatanga – Ghana

Intern September 2020 – December 2020

- Designed a one-story store unit and a two-story school block using Autodesk AutoCAD Architecture and Autodesk Revit software.
- Inspected construction sites with engineers frequently to ensure construction was done according to designs.

Professional Development and Training

Workshops

Engineering Week Celebration, Ghana Institute of Engineers (GhIE-UENR) Sunyani, Ghana

Role: Instructor September 2022

- Conducted hands-on training sessions on Revit and AutoCAD for engineering students as part of the Skills Development Hub, focusing on building modeling fundamentals and drafting workflows.

Civil Engineering Design Workshop – Department of Civil Engineering, UENR Sunyani, Ghana

Role: Instructor May 2023

- Delivered practical training in Revit and Prota-Structures, guiding participants through integrated design workflows for civil and structural engineering projects.

Conferences and Seminars

- 11th International Conference on Construction Engineering and Project Management (ICCEPM 2025), Jeju, South Korea, (November 2025). ICCEPM 2025
- 2025 Ghana Students in Korea & Associates (GHASKA) Innovative Conference, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, 16 August 2025. GHASKA
- Smart Construction IT & Safety Seminar (2025 년 스마트 건설 IT & 안전 세미나) — Attendee, Chungbuk National University, Cheongju, 5 Aug 2025. Leadership Experience.
- 2021 Hult Prize Challenge, Regional Finals, Nairobi Impact Summit, April 2021. Regional Finals Competitor. HultPrize 2021

Leadership Experience

Academic Board Committee, Ghana Institute of Engineers – UENR Chapter Sunyani– Ghana

1st Deputy Chairperson GHIE January 2022 – September 2022

- Assisted in the preparation of a constitution to govern and manage the affairs of the organization.
- Organized an academic quiz competition as part of the GhIE week celebration.

Academic Board, Students' Representative Council – UENR Sunyani– Ghana

Civil Engineering Department Representative CESA January 2021 – September 2021

- Promoted the implementation of the mid-semester examination week.
- Educated students on examination malpractices using info posters and class-to-class sensitization.

Membership and Affiliations

- Professional Engineer – Institute of Engineering and Technology, Ghana (IET - GH), PE 05439

Skills & Interests

- **Programming and Deep Learning** for automated point cloud processing in Scan-to-BIM using Python, PyTorch, and Open3D.
- **3D Vision & Point Cloud Processing** for visualizing, registering, and preprocessing point cloud datasets using CloudCompare, Open3D, and MATLAB.
- **3D Modeling & BIM Design** for creating and modifying architectural and structural models with Blender, Autodesk Revit, AutoCAD, and SketchUp**.**
- **3D Simulation & Robotics** for designing and validating robotic systems using ROS2, Gazebo, and Blender-based simulation environments.
- **Advanced Data Analysis** using SPSS, Excel, ANSYS, ETABS, MIDAS Civil, and Python.
- **Creative Design and Research Infographics** using Adobe Photoshop/Premiere Pro, PowerPoint.
- **Laboratory & Field Equipment:** Operating Terrestrial and Mobile Laser Scanners (TLS/MLS), UAVs (DJI Mavic series), and Unitree Go2 quadruped robots for data acquisition and testing.
- **Language:** English (fluent) and French (basic)

References

Dr. Kim Minkoo, Associate Professor
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